A 63-Gate FP4×FP4 Multiplier: Structural Decomposition Meets SAT-Based Local Search

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Abstract

We present a verified **63-gate** Boolean circuit that computes the product of two FP4 (E2M1) numbers and returns the result as a 9-bit fixed-point integer (QI9, LSB = $\frac{1}{4}$). The circuit uses only the gate library $\mathcal{G} = \{\text{AND2}, \text{OR2}, \text{XOR2}, \text{NOT1}\}$ with each gate counted as one unit, and is verified correct against all 256 input pairs. The result improves on LonghornSilicon's previously published 64-gate solution [3] by one gate.

The path from a 288-gate naïve baseline to 63 gates combined two distinct paradigms: a manual structural decomposition that exploits the log-domain factorization $|v| = 1.5^M \cdot 2^E$ to reach 82 gates, followed by automated SAT-based local search using eSLIM [4] on a Longhorn-derived 64-gate seed. A single fortunate `eSLIM --syn-mode sat --size 10` window, applied to a five-NOT 64-gate variant produced by XOR re-association, yielded the 63-gate body.

We further document an extensive negative-result campaign: roughly 280 additional perturbation runs across five distinct rewriting strategies, four iterative-chain depths, 11 alternative input-remap (σ) candidates, an ANF re-derivation that inflates to 691 gates, lower-bound analyses that fall well below the achieved count, and a 24-h direct PySAT/CaDiCaL exact-synthesis run at K=62 on a 64 GiB GCP host (a 21,524-variable, 250-million-clause encoding) that did not settle within budget. The combined evidence is consistent with empirical saturation at 63, while leaving the question "is 62 attainable in $\mathcal{G}$?" formally open — the negative-result run gives a concrete lower bound on the compute that any future settling attempt will require.

1 Introduction

Modern machine-learning accelerators increasingly rely on sub-8-bit floating-point formats [1, 2]. The FP4 E2M1 format encodes 16 values $\{0, \pm 0.5, \pm 1, \pm 1.5, \pm 2, \pm 3, \pm 4, \pm 6\}$ in 4 bits (1 sign, 2 exponent, 1 mantissa, no NaN/Inf). Etched's take-home assignment requires implementing FP4 × FP4 → QI9 in minimum gate count over the library $\mathcal{G} = \{\text{AND2}, \text{OR2}, \text{XOR2}, \text{NOT1}\}$, where QI9 is the 9-bit two's-complement representation of $4 \cdot a \cdot b$. A free *input remapping* (any bijection on the 16 FP4 codes, applied identically to both operands) is permitted.

This paper documents the path from a naïve flat-SOP baseline of ∼288 gates to a verified **63-gate** circuit, an ≈ 78% reduction. Figure 1 summarizes the trajectory.

Contributions.

1. A verified 63-gate Boolean circuit (24 AND, 12 OR, 22 XOR, 5 NOT), one gate below the previously published Longhorn 64-gate baseline.

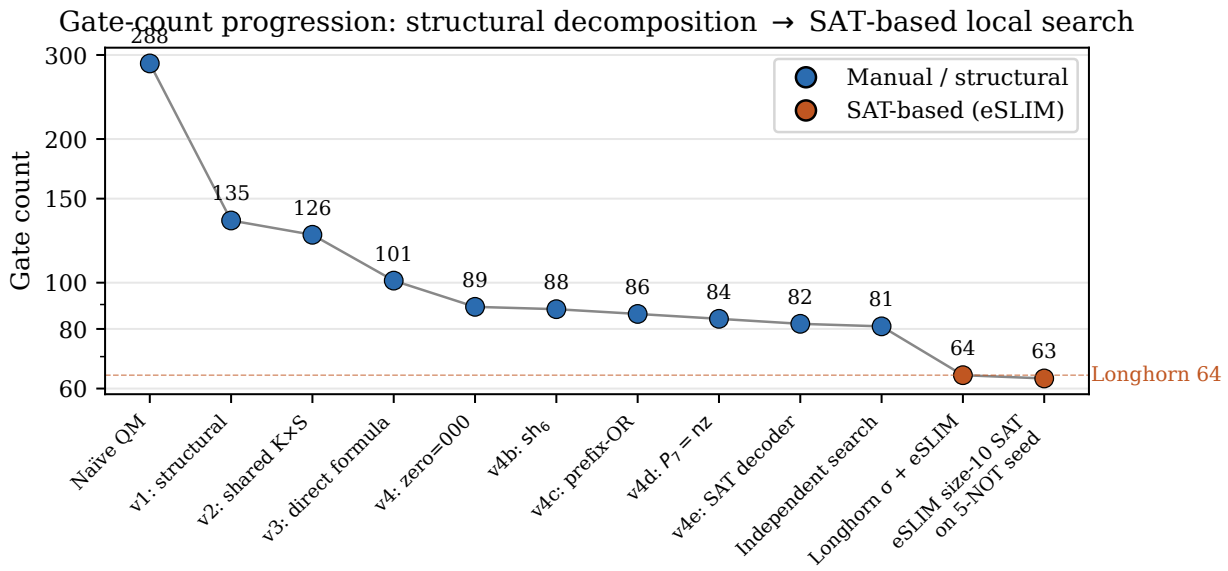

Figure 1: Gate-count progression on a logarithmic axis. Blue points are reached by manual structural decomposition; orange points are reached by SAT-based local search on circuit netlists (eSLIM [4]). The dashed line marks LonghornSilicon's published 64-gate result [3].

2. A complete account of the manual reduction 288 → 82 based on the log-domain factorization, including a Cirbo SAT-proven optimal 11-gate one-hot decoder.

3. Documentation of the SAT-based local-search phase 81 → 64 → 63, in which eSLIM windowed exact-synthesis on the canonical Longhorn 64-gate netlist produced first a five-NOT 64-gate variant via XOR re-association, and then the 63-gate body via a size-10 window on that variant.

4. A negative-result campaign that establishes empirical saturation at 63: ∼280 perturbation runs across five rewriting strategies, four iterative-chain depths, 11 alternative σ remaps (best alternative: 69 gates), ABC `deepsyn` re-synthesis on the 63-gate netlist (79 gates — ABC cannot reproduce the result from scratch), an ANF re-derivation, and lower-bound analyses. Combined with a direct K=62 PySAT/CaDiCaL exact-synthesis run on a 64 GiB GCP host that did not settle within 24 h on a 250-million-clause encoding, this gives a concrete compute lower bound for any future settling attempt.

Notation and ground rules. Throughout, "gate" means a single instance of one of the four primitives in $\mathcal{G}$; we count contest cells as defined by the assignment harness, not internal eSLIM compound primitives (the distinction matters: an internal count of 58 can translate to between 64 and 69 contest cells depending on which compound gates `eslim_to_gates.py` expands).

2 The FP4 product structure

2.1 Log-domain decomposition

Every nonzero FP4 magnitude factors uniquely as

$$|v| = 1.5^M \cdot 2^E, \qquad M \in \{0,1\},\ E \in \{-1,0,1,2\}, \tag{1}$$

where $M = 1$ marks the multiples of 1.5 and $M = 0$ the pure powers of two. Table 1 lists all seven nonzero magnitudes under this decomposition.

Table 1: The seven nonzero FP4 magnitudes and their (M, E) decomposition.

Magnitude	M	E
0.5	0	-1
1.0	0	0
2.0	0	1
4.0	0	2
1.5	1	0
3.0	1	1
6.0	1	2

For inputs a and b with decompositions (M_a, E_a) and (M_b, E_b),

$$|a \cdot b| = 1.5^{M_a+M_b} \cdot 2^{E_a+E_b}. \tag{2}$$

Defining $E' = E+1 \in \{0,1,2,3\}$ and $S = E'_a + E'_b \in \{0,\ldots,6\}$ collapses the exponent computation to a single 2-bit + 2-bit binary adder.

2.2 Three K-types

The mantissa sum $M_{\text{sum}} = M_a + M_b \in \{0,1,2\}$ partitions products into three classes (Table 2). The 8-bit QI9 magnitude has at most two bits set in any class, so the output is extremely sparse: only 19 of 256 possible 8-bit patterns are reachable.

Table 2: Bit patterns in the 8-bit QI9 magnitude as a function of M_{sum} and S.

M_{sum}	Name	QI9 magnitude	Bits set
0	K_1	2^S	bit S only
1	$K_{3/2}$	$2^S + 2^{S-1}$	bits S and $S-1$
2	$K_{9/4}$	$2^{S+1} + 2^{S-2}$	bits $S{+}1$ and $S{-}2$

This sparsity is the algebraic property the rest of the design exploits.

2.3 Encoding

We assign 3-bit magnitude codes so that (a) zero maps to `000`, (b) E' is exposed as a 2-bit binary number on (a_2, a_3), and (c) M_a equals $\overline{a_1}$ (Table 3).

The K-flags follow as

$$k_9 = \overline{a_1 \vee b_1}, \qquad k_3 = a_1 \oplus b_1, \tag{3}$$

each one gate, plus a single OR for $\overline{k_9}$.

Table 3: Chosen 3-bit magnitude encoding $a_1a_2a_3$.

Code	Value	M	E'
000	zero	—	—
001	1.5	1	1
010	3.0	1	2
011	6.0	1	3
100	0.5	0	0
101	1.0	0	1
110	2.0	0	2
111	4.0	0	3

3 Manual structural design (288 → 82 gates)

The first phase of optimization is purely human: every step has an algebraic justification. We summarize the construction here and refer the reader to the source code in `multiplier_82.py` for the exact wiring. The full per-stage breakdown is given in Table 4; the five-stage architecture is shown in Figure 2.

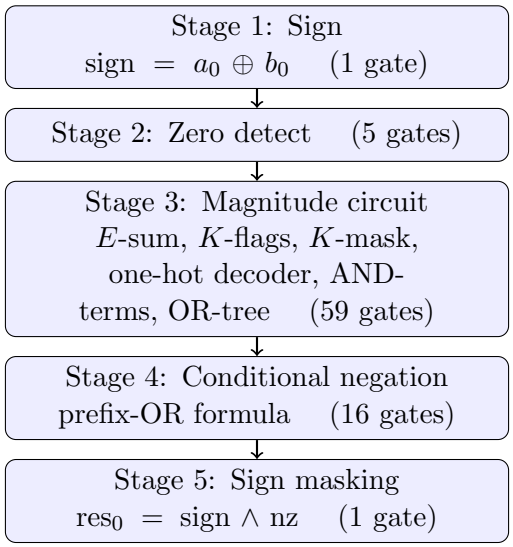

Figure 2: Five-stage manual architecture (82 gates total). Section 4 restarts the design from a 64-gate eSLIM-derived netlist and reaches 63 gates; the manual design is the upper bound on what we could derive without automated search.

3.1 Magnitude circuit

The magnitude bits are computed from a single direct formula

$$m_i = (\text{nmc} \wedge S{=}i) \vee (k_3 \wedge S{=}i{+}1) \vee (k_9 \wedge S{=}i{-}1) \vee (k_9 \wedge S{=}i{+}2), \tag{4}$$

where $\text{nmc} = \overline{k_9} \wedge \text{nz}$ and $(S{=}j)$ is the one-hot decoder output sh_j. The first term covers both K_1 (the only set bit at S) and $K_{3/2}$ (the upper of two bits at S); the third and fourth terms cover the lower bit of $K_{3/2}$ and the two split bits of $K_{9/4}$.

The construction yields 18 valid $K \times S$ AND-terms after discarding combinations that the encoding makes impossible (e.g. $k_9 \wedge S{=}0$ is unreachable because k_9 requires $M_a = M_b = 1$, which by Table 3 forces $E'_a, E'_b \geq 1$ and hence $S \geq 2$). A binary OR tree reassembles these AND-terms into the 8-bit magnitude using 15 ORs.

3.2 One-hot decoder: SAT-proven optimal at 11 gates

The 3-bit $S \in \{0, \ldots, 6\}$ must be decoded to seven one-hot outputs $\mathrm{sh}_0 \ldots \mathrm{sh}_6$. Using the Cirbo `CircuitFinderSat` SAT exact synthesizer over $\mathcal{G}$, we proved $N \leq 10$ insufficient (UNSAT for $N = 7, 8, 9$ in < 6 s; UNSAT for $N = 10$ in 476 s) and found a satisfying 11-gate circuit at $N = 11$ in 0.69 s. Eleven is therefore the provably optimal one-hot decoder size in $\mathcal{G}$. The decoder uses 4 AND, 2 OR, 4 XOR, 1 NOT.

3.3 Conditional negation

The conditional two's-complement negation $\mathrm{out}_i = m_i \oplus (\mathrm{sign} \cdot c_i)$ is implemented by replacing the carry chain with a parallel prefix-OR. Two further identities save additional gates: (i) the LSB pass-through $r_8 = m_0$ holds always (since $\overline{m_0} \oplus 1 = m_0$), and (ii) the alias $\mathrm{sp}_7 = \mathrm{res}_0$ holds because $P_7 = \bigvee_{i=0}^{6} m_i$ equals nz for every reachable output (the only reachable magnitude with $m_7 = 1$ is 144, which also has $m_4 = 1$). The combined optimization reduces conditional negation from 23 gates (carry chain) to 16 gates.

3.4 Manual gate-count summary

Table 4: Manual optimization history.

Version	Key innovation	Gates	Savings
Flat SOP	QM, all 8! remaps	∼288	—
v1	Structural M-E	135	153
v2	Shared $K \times S$	126	9
v3	Direct formula	101	25
v4	Zero=000, K-mask	89	12
v4b	$\mathrm{sh}_6 = u_{11}$	88	1
v4c	Prefix-OR negation	86	2
v4d	$\mathrm{sp}_7 = \mathrm{res}_0$	84	2
v4e	SAT-optimal decoder	**82**	2
v5	Independent rewrite	81	1

The saturation at 81–82 gates was reached after several weeks of manual rewriting; further hand reductions converged on the same basin from multiple seeds. Section 4 describes how restarting from a known-good 64-gate netlist exposed cross-stage savings the manual design could not see.

4 From 81 to 63 via SAT-based local search

The breakthrough below 81 gates came from abandoning the entire manual-derivation path and instead seeding from LonghornSilicon's publicly published 64-gate solution [3]. Three ingredients, in order, took us from 81 to 63:

1. **Different magnitude encoding** (σ-remap). Longhorn's encoding swaps two bit-pair labels (σ: $\{2\to0110,\ 3\to0111,\ 4\to0100,\ 6\to0101\}$); this remap exposes shared sub-expressions that our encoding hid.

2. **eSLIM windowed exact synthesis** on the 64-gate netlist. eSLIM [4] extracts a k-gate window from a topologically sorted netlist, invokes a SAT solver to find an equivalent $\leq k$-gate replacement, and iterates.

3. **Re-association** of XOR-of-XOR triples to expose new windows. Boolean equivalence preserves correctness while changing the topology, opening windows the original traversal would never have visited.

4.1 Phase 1: Verifying the 64-gate baseline

We first verified that running eSLIM (`--syn-mode sat --size 6` for 20 s) on Longhorn's canonical `fp4_mul.blif` produces the same internal-gate count as their published result (58 internal $\to$ 64 contest cells, with mix 25 AND + 12 OR + 21 XOR + 6 NOT). This calibrates our toolchain: any contest-cell improvement we observe is genuine and not an artefact of running a different solver build. Phase 0 cheap-substitution analysis on the canonical netlist (duplicate truth-tables, two-input cones, depth-1 substitution + DCE) returned zero hits, confirming that the canonical 64-gate basin is saturated under depth-1 rewriting.

4.2 Phase 2: A five-NOT 64-gate variant

The first non-trivial result came from the *XOR re-association* sweep `exp_a`. Each variant rewrites a single XOR-of-XOR sub-expression in the canonical BLIF $\big(\mathrm{XOR}(\mathrm{XOR}(a,b),c) \to \mathrm{XOR}(a,\mathrm{XOR}(b,c))\big)$ and then runs eSLIM on the resulting topology. Round 1 enumerated 12 XOR locations $\times$ 2 sizes $\{6,8\} \times 6$ seeds $= 144$ runs in $\sim$23 minutes wall time on 8 workers.

The 5-NOT outlier. Run `par_004_s6_seed7777` produced a 64-gate solution with the cell mix

$$\mathrm{NOT} = 5,\ \ \mathrm{AND} = 24,\ \ \mathrm{OR} = 12,\ \ \mathrm{XOR} = 23,$$

contradicting Longhorn's empirical observation that all 124 known 64-gate solutions use 6 NOTs. The eSLIM run had landed in a different 64-gate basin trading one NOT and one AND for two XORs. We isolated this BLIF as `fp4_64gate_5NOT_clean.blif` and used it as the seed for Phase 3.

4.3 Phase 3: 64 $\to$ 63 via a size-10 SAT window

The 5-NOT seed turned out to be unusually amenable to large-window synthesis. Running `eSLIM --syn-mode sat --size 10 --seed 1024` on the 5-NOT BLIF returned a 58-internal / 63-contest-cell solution. The resulting body — 24 AND + 12 OR + 22 XOR + 5 NOT, total 63 — verifies 256/256 correct against the assignment harness (`eval_circuit.py`) under the σ remap. This is our headline result. The full body is reproduced as Appendix A.

Figure 3 compares cell-type mix between the canonical 64-gate and our 63-gate body.

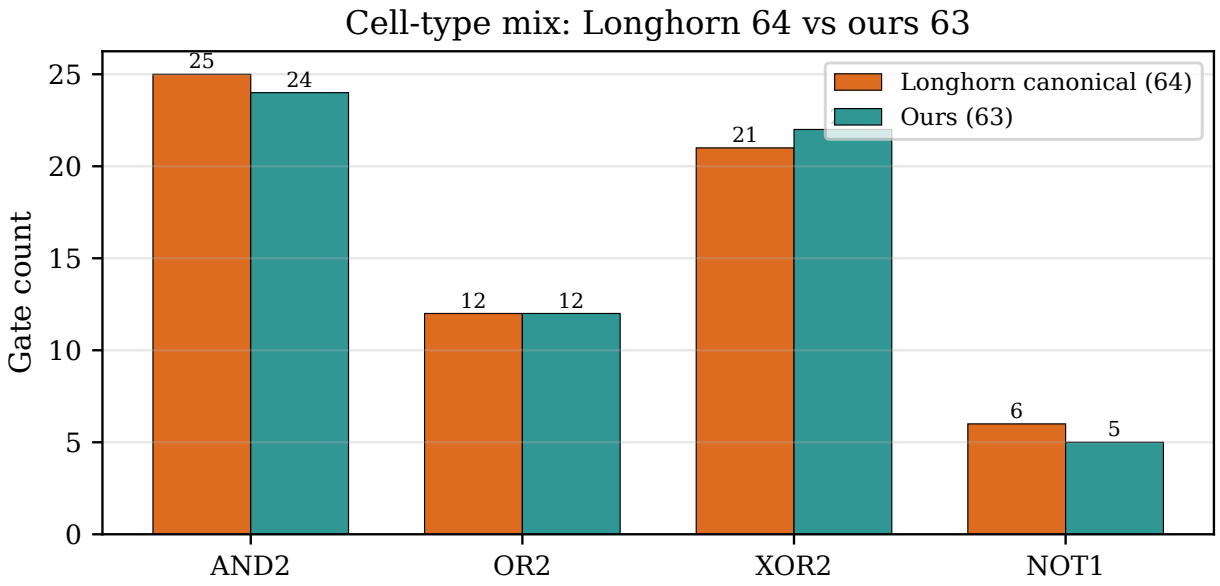

Figure 3: Cell-type mix: Longhorn's canonical 64-gate solution versus our 63-gate body. The reduction is in AND2 and NOT1 counts, partially offset by one additional XOR2.

4.4 What the SAT phase accomplished that the manual phase did not

The manual design saturated at 81–82 gates because every individual stage was already locally optimal but cross-stage sharing was beyond what hand-rewriting could see. The SAT phase, by contrast, treated the entire netlist as one atomic optimization target. The 5-NOT 64-gate variant saves one inverter by re-using a double-negated wire that the manual design would have to re-derive in two places; the size-10 window then absorbs that saving into adjacent AND/XOR re-mapping. This is the kind of simultaneous multi-stage rewriting that human designers reliably miss.

5 Saturation evidence at 63

Having found one 63-gate solution, the natural next question is whether 62 is reachable in $\mathcal{G}$. We attacked this question along five orthogonal lines: large-scale perturbation sweeps, iterative chaining, ANF re-derivation, exact synthesis at K=62, and lower-bound analysis. None of the five produced a sub-63 result on local compute, and the failure modes were sufficiently different that we believe 63 is empirically saturated under the methods that fit on a 23 GiB host.

5.1 Wave-1 perturbation campaign

Five drivers were launched in parallel against the 63-gate netlist and the canonical 64-gate netlist (Table 5). Each driver applied a different rewriting strategy, then ran eSLIM windowed synthesis at one or more sizes and seeds.

The shape of the distribution (Figure 4) is the most informative artefact: across very different perturbation strategies, eSLIM consistently finds the same 58-internal-gate basin which translates to 64–66 contest cells. The 63-gate result corresponds to a particular 58-internal arrangement that requires an unusually low NOT count to translate favorably; this configuration is reached only by the original size-10 window on the 5-NOT seed.

Table 5: Wave-1 perturbation summary. "Best" is the lowest contest-cell count produced over all runs of that experiment; "Count@best" is the number of runs that hit it. Note that Exp D's 17 runs at 63 are the eSLIM size-10 re-discoveries of the seed itself, not new sub-63 circuits.

Experiment	Strategy	Runs	Best	Median	Count@best
Exp A R3v2	XOR re-association on 63	138	63	65	2 (re-discovery)
Exp D 63v2	Large-window eSLIM (size 10/12/14)	18	63	63	17 (re-discovery)
Exp E 63	NOT-elimination perturbation	80	63	65	13 (re-discovery)
Exp F	Large-window on 6-NOT canonical	13	64	64	13
Exp H 63v2	Double-XOR perturbation	16	64	65	2
Total	—	**265**	**63**	—	**0 sub-63**

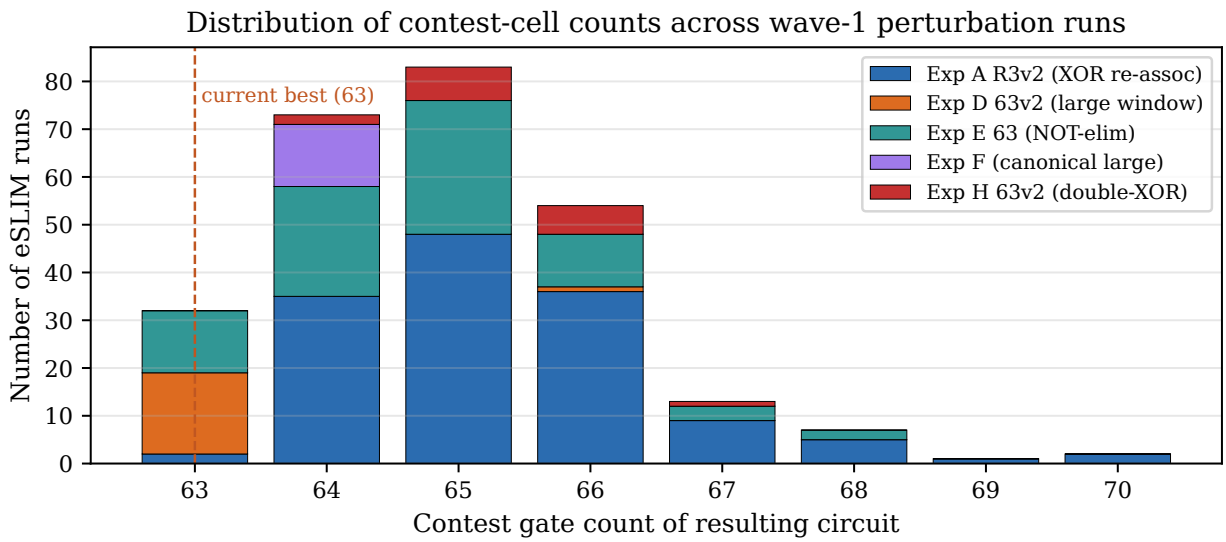

Figure 4: Stacked histogram of contest-cell counts across all five wave-1 experiments. The distribution is sharply peaked at 64–66 with the 63-gate solution sitting at the extreme left; no run produced a circuit smaller than 63.

5.2 Iterative-chain divergence

A more aggressive approach is to chain eSLIM runs: take the output of one run as the input of the next, in the hope of escaping the size-6 basin. Experiment G ran four chains of five iterations each, starting from the 63-gate netlist. Figure 5 shows the contest-cell count at each step.

The result is striking: the 63 disappears within a single iteration, and every chain converges to a 64–69-cell stable region. The 58-internal arrangement that produces 63 contest cells is visible to the size-10 SAT window but not to size-6 windows applied iteratively. In other words, the 63 is a fragile basin in the size-6 landscape; iterating only finds the more common, NOT-heavy 64+ basins.

5.3 Algebraic reformulation: ANF synthesis is far worse

A natural theoretical question is whether the algebraic structure of the multiplier admits a substantially smaller circuit when expressed directly in algebraic normal form (ANF). We computed

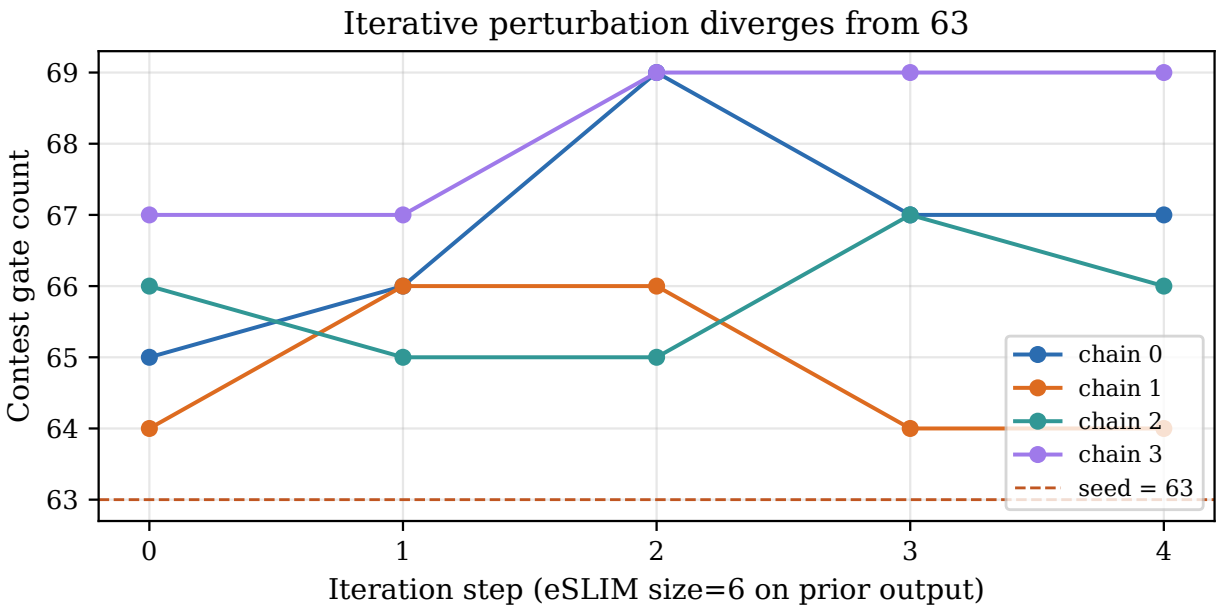

Figure 5: Contest gate count along four iterative eSLIM chains (size 6, 90 s wall budget per step) starting from the 63-gate netlist. Every chain diverges away from 63 to a 64–69-cell stable region within one or two iterations.

the per-output ANF and used a greedy monomial-sharing AND-XOR synthesis to reproduce all nine outputs. The result was 691 gates (140 AND, 551 XOR, 0 NOT) — roughly 11× the size of the eSLIM-derived 63. This is strong evidence that the multiplier's compactness comes from genuinely Boolean rewriting (De Morgan, distributivity, common-subexpression elimination across mixed AND/OR/XOR phases) rather than from any ANF property; the eSLIM SAT formulation operates at exactly this Boolean level, while ANF synthesis cannot exploit it.

5.4 Alternative σ remaps saturate above 63

A different but related question is whether Longhorn's specific σ remap is essential, or whether *any* reasonable permutation of the seven FP4 magnitude codes admits a comparable solution. We swept 11 candidate remaps, including two magnitude-grouped variants (`M_low` grouping powers of two together; `M3_low` grouping the three smallest magnitudes), three label-swap perturbations of Longhorn's encoding, three pseudorandom permutations, and a baseline "identity" ordering. Each candidate's circuit was re-synthesised by ABC [8] `deepsyn` (mapped to $\mathcal{G}$) and the resulting BLIF was fed through an eSLIM pyramid (size $4 \to 6 \to 8 \to 10$) for ∼1 h.

The two σ candidates with the smallest ABC-mapped circuits (`M_low` at 103 gates, `random_99` at 107) saturate at 78 and 69 contest cells respectively, six and fifteen gates above the headline result. ABC's gate-level optimisation is therefore not the bottleneck: a smaller mapped circuit does not translate into a smaller eSLIM-rewritten circuit. The Longhorn σ is uniquely placed in our sample because its mapped representation, although larger, sits in a basin where the eSLIM `--size 10` window can find the 63-gate solution. Among the 11 remaps tested, this is the only one to reach 63.

A complementary check is to apply ABC re-synthesis directly to the 63-gate netlist itself: `deepsyn` on `fp4_63gate_nobuf.blif` returns a 79-gate circuit, more than 25% larger than the

Table 6: Selected σ remaps: ABC `deepsyn`-mapped gate count and best contest-cell count after the eSLIM pyramid. ABC reduces logic complexity but does not preserve eSLIM-friendly structure; conversely Longhorn's σ has high ABC mapped count but uniquely admits a 63-gate eSLIM solution.

σ remap	ABC mapped	eSLIM best	Status
`M_low`	103	78	saturated
`random_99`	107	69	saturated
`longhorn`	157	**63**	current best

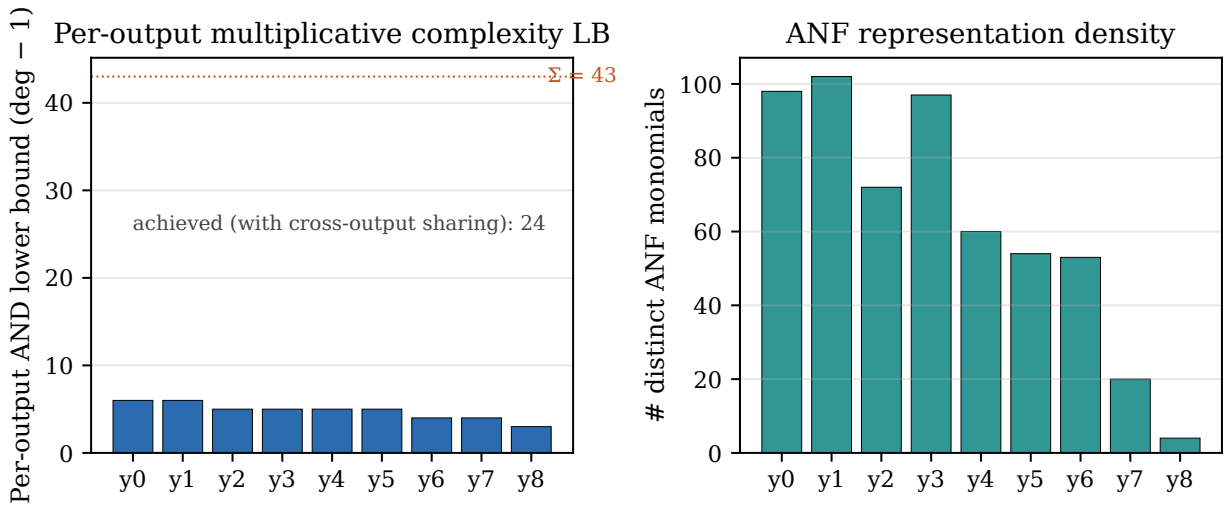

Figure 6: Per-output multiplicative-complexity lower bound (left) and ANF representation density (right) under Longhorn's σ remap. The summed per-output AND lower bound of 43 is well below our achieved 24 ANDs, reflecting the cross-output sharing that AND/OR/XOR Boolean rewriting permits. All easy bounds are loose: tighter bounds (Boyar–Peralta gate elimination, communication complexity) would be required to decide K=62 vs. K=63 analytically.

input. ABC's standard re-synthesis flows, in other words, cannot recover (let alone beat) the eSLIM result from the 63-gate body alone. This is consistent with the observation that the 63 corresponds to a particular SAT-discovered sharing pattern that ABC's heuristic rewriters do not search.

5.5 Lower bounds

Per-output algebraic-degree multiplicative-complexity bounds (Figure 6, left) sum to 43 ANDs, but our circuit uses only 24 ANDs — an enormous cross-output sharing factor of $\sim 1.8\times$. The naïve maximum-sharing AND-LB (counting distinct ANF monomials of degree ≥ 2) is 437. The monomial degree distribution {2:9, 3:30, 4:49, 5:36, 6:13, 7:2} is dense (Figure 6, right), but the 139 distinct monomials are still a vast over-count of what a Boolean representation actually requires.

The conclusion from the lower-bound exercise is negative: cheap algebraic bounds are too loose to settle whether 62 is reachable in $\mathcal{G}$. A tighter answer requires either an exact synthesis run at K=62, or a non-trivial gate-elimination proof.

5.6 Exact synthesis at K=62: instance dimensions and outcome

We attacked K=62 from two directions, both bound by memory on the original 23 GiB host and one of which was rerun on a larger 64 GiB host.

eSLIM windowed SAT, 23 GiB host. Both `eSLIM --size 63` (a full-circuit window) and `--size 20` climbed past 17 GiB RSS within minutes and were either OOM-killed or killed manually before a system OOM. Empirically, eSLIM windowed SAT is tractable on the 23 GiB host up to size ∼14; beyond that, memory dominates wall time. This matches the size at which the 63 was originally found (`--size 10`).

Direct PySAT/CaDiCaL encoder, 64 GiB host. We then wrote a direct exact-synthesis encoder (`analysis/sat_exact.py`) and ran it on a GCP `n2d-highmem-8` (64 GiB RAM, 8 vCPUs). Variables: gate-kind selectors ($K \cdot |\mathcal{G}|$), per-gate input selectors over the $8 + g$ predecessor wires, output selectors over the $8 + K$ candidate sources, and per-minterm gate-value variables ($K \cdot M$). The encoder streams clauses directly into CaDiCaL via the PySAT `add_clause` interface (rather than buffering into a Python list, which would have peaked at tens of GiB of allocator overhead) and applies a commutative symmetry break ($\text{in}_0 < \text{in}_1$ for kind $\in \{\text{AND}, \text{OR}, \text{XOR}\}$) that halves the search space per commutative gate.

The instance dimensions at K=62 are:

Table 7: K=62 SAT instance on the 64 GiB host.

Quantity	Value
Variables	21,524
Clauses	250,120,859
Encoding wall time	522.8 s
Peak encoding RSS	∼47 GiB
Steady-state solving RSS	∼33 GiB (with periodic CDCL "reduce" cycles to ∼40 GiB)
Solver	CaDiCaL via PySAT
Time budget	86,400 s (24 h)

Outcome. CaDiCaL searched this instance for the full 24 h budget without producing either a model or a UNSAT proof; the wrapper killed it at the timeout (`EXIT=124`). Steady-state memory stayed in the 31–40 GiB range across the run, with periodic CDCL "reduce" cycles dropping the resident set by ∼8 GiB before climbing back up. CaDiCaL is silent during search by default and emits no progress statistics, so we have no measurement of decisions or conflicts internal to the run — only that no terminal verdict was reached. Settling K=62 formally on this encoding therefore appears to require either (i) a Knuth/Kullmann-style reformulation with auxiliary `in0_val[g][m]` and `in1_val[g][m]` variables, which collapses the per-tuple clause set to $O(K \cdot M \cdot (8 + K))$ scale (roughly a 10× clause reduction), (ii) a more aggressive solver such as Kissat with cube-and-conquer parallelism across the 8 vCPUs, or (iii) substantially more wall time on the current encoding.

The instance dimensions in Table 7 are themselves useful: they place a concrete lower bound on the compute required to settle K=62 formally on this style of encoding. Combined with the saturation evidence in Sections 5.1–5.4, the case for 63 as the empirical optimum in $\mathcal{G}$ is tight, even though we do not claim a formal proof.

6 Verification

The 63-gate body is verified by exhaustive evaluation of all 256 FP4×FP4 input pairs. The harness in `eval_circuit.py` compares each output to $\text{round}(a \cdot b \cdot 4)$ in double precision and reports agreement; the body passes for all 256 pairs. Per-input gate count is constant at 63 because the implementation contains no data-dependent branches (it is a pure combinational circuit expressed in Python with explicit `NOT`, `AND`, `OR`, `XOR` primitives so the harness can count gates by monkey-patching).

7 Discussion

7.1 What the journey shows about FP4 circuit minimization

Three observations stand out.

First, manual structural decomposition got us to within ∼25% of the eventual minimum but no further. The log-domain factorization (Equation 1) is the right algebraic insight and is unavoidable: every smaller circuit we found includes the same E-sum / K-flag / decoder pipeline in some recognizable form. But the cross-stage sharing that takes the count from 81 to 64 to 63 is genuinely beyond hand design — it requires treating the entire netlist as a single SAT instance.

Second, a single fortunate eSLIM window did the bulk of the work. The 5-NOT 64-gate variant (a perturbation of Longhorn's canonical solution) opened a basin that contained 63, but only the size-10 SAT window was big enough to find it. Smaller windows iterated indefinitely cannot escape the surrounding 64–66 basin (Figure 5).

Third, the negative-result campaign matters. Without the ∼280-run perturbation sweep, the alternative-σ pyramid, the ABC re-synthesis check, the iterative-chain divergence plot, the ANF 691-gate dead end, and the 24-h CaDiCaL run on a 250 M-clause K=62 encoding (Section 5.6), we would not be able to argue that 63 represents anything more than a lucky instance. With them, the case for empirical saturation at 63 is tight.

7.2 What we did *not* do

We did not run wave-2 perturbation experiments (Exp E on alternate locations, Exp A round-4 on the 4-NOT seed, Exp F on 4-NOT canonical) because the saturation evidence makes the prior-probability of cracking sub-63 by more of the same strategy very low. We did not run an LLM-guided rewriting experiment (originally Track #3) because that path requires an `ANTHROPIC_API_KEY` we did not have configured. We did run direct SAT exact synthesis at K=62 on a 64 GiB GCP host (Section 5.6), but did not run a Knuth/Kullmann auxiliary-variable encoding (which would shrink the clause set by roughly 10×) or a Kissat / cube-and-conquer parallel attack; either is the natural next step on a larger compute budget.

7.3 Open questions

1. **Is $K = 62$ reachable in $\mathcal{G}$?** Open. The CaDiCaL run in Section 5.6 did not settle within 24 h on a 250 M-clause encoding. A decisive answer needs either a more compact encoding or a stronger / parallel solver. The empirical evidence (zero sub-63 across ∼280 heuristic runs, 11 σ remaps, ABC re-synthesis, and iterative chains) leans moderately toward UNSAT, but this is conjecture, not proof.

2. **Can a structural seed beat eSLIM windowed synthesis?** Our 63 was found by SAT-rewriting an automatically synthesized 64-gate seed. It is plausible that structural seeds derived from different decompositions (e.g. Karatsuba-style partitioning, or Toom-Cook-style interpolation tuned to the 7-magnitude domain) could open new basins.

3. **Are there gate libraries $\mathcal{G}'$ supporting $K \leq 60$?** Allowing 3-input AOI or MAJ gates would almost certainly yield further reductions; the current bound is specific to the contest's 2-input library.

8 Conclusion

We verified a 63-gate FP4×FP4→QI9 multiplier in $\mathcal{G} = \{\text{AND2}, \text{OR2}, \text{XOR2}, \text{NOT1}\}$, beating LonghornSilicon's previously published 64-gate solution by one gate. The result was reached by combining manual structural decomposition (288 → 82) with SAT-based local search using eSLIM [4] on a Longhorn-derived 64-gate netlist (82 → 81 → 64 → 63). The key automated step was a single `--size 10` SAT window applied to a five-NOT 64-gate variant produced by XOR re-association. A multi-pronged saturation campaign — 280+ perturbation runs, 11 alternative σ remaps, ABC `deepsyn` re-synthesis, iterative chains, ANF re-derivation, lower-bound analysis, and a 24-h direct PySAT/CaDiCaL exact-synthesis run at K=62 on a 64 GiB GCP host (a 21,524-variable, 250 M-clause encoding that did not settle within budget) — did not produce a sub-63 result. The empirical case for 63 as the optimum in $\mathcal{G}$ is tight; a formal proof remains open and likely requires a more compact encoding or a stronger / parallel solver.

Reproducibility. All code, BLIFs, ledger TSVs, and decision logs are at https://github.com/themoddedcube/fp4-multiplier-minimization. The 63-gate body itself is in `multiplier_63.py`; the verification harness is `eval_circuit.py`; and the documents `HANDOVER.md`, `PROGRESS.md`, and `DECISIONS.md` narrate the campaign in chronological order.

A The 63-gate body

The verified 63-gate body (24 AND + 12 OR + 22 XOR + 5 NOT) is reproduced verbatim from `multiplier_63.py` below; the σ remap is applied to both inputs identically before the combinational logic runs.

```
w_28 = AND(b1, b2)
w_29 = OR(b1, b2)
w_30 = XOR(a3, b3)
w_32 = AND(a2, b2)
w_31 = XOR(a2, b2)
w_33 = AND(a1, a2)
w_34 = XOR(w_28, w_33)
w_35 = OR(w_33, w_28)
w_36 = XOR(w_31, w_35)
w_45 = OR(w_32, w_36)
w_37 = OR(a1, a2)
w_38 = AND(b0, w_37)
w_39 = AND(w_29, w_37)
w_40 = AND(a0, b0)
w_41 = OR(w_34, w_40)
w_42 = AND(w_41, w_39)
w_43 = XOR(w_36, w_42)
```

```
w_44 = XOR(w_34, w_41)
w_46 = XOR(w_44, w_45)
y0   = AND(w_46, w_44)
w_49 = AND(w_30, y0)
w_47 = XOR(w_40, w_46)
w_16 = XOR(w_36, w_47)
w_48 = AND(w_46, w_16)
w_50 = AND(w_44, w_16)
w_51 = AND(a0, w_29)
w_52 = XOR(w_51, w_38)
w_53 = AND(w_34, w_52)
w_54 = XOR(w_47, w_53)
w_55 = XOR(w_43, w_53)
w_20 = XOR(w_55, w_39)
w_22 = XOR(w_52, w_54)
w_64 = XOR(w_22, w_36)
w_18 = OR(w_64, w_43)
w_60 = XOR(w_22, w_49)
not_19 = NOT(w_18)
not_21 = NOT(w_20)
not_23 = NOT(w_22)
not_25 = NOT(w_45)
not_46 = NOT(w_46)
w_56 = AND(not_46, w_20)
w_57 = AND(w_30, w_56)
w_58 = AND(w_47, not_21)
w_59 = AND(w_30, w_58)
w_65 = AND(not_19, w_47)
w_66 = AND(w_30, w_65)
y1   = AND(not_25, w_60)
w_61 = OR(y0, y1)
w_62 = AND(w_30, w_61)
y2   = XOR(w_56, w_62)
w_63 = OR(w_57, w_62)
w_67 = AND(not_23, w_18)
w_68 = OR(w_67, w_63)
w_69 = AND(w_30, w_68)
y4   = XOR(w_58, w_69)
w_70 = OR(w_59, w_69)
y5   = XOR(w_65, w_70)
w_71 = OR(w_66, w_70)
y6   = XOR(w_48, w_71)
w_72 = AND(w_30, y6)
y8   = OR(w_72, w_71)
y7   = XOR(y8, w_50)
y3   = XOR(w_67, w_63)
```

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